s7_directional_aggressive

equity strategy | family: equity-aggressive | status: research | derived from `s7_directional`

What this strategy does

s7_directional re-sized for return: base unit 0.006, 6x leverage cap

Why it should work. Let net exposure float with the signal; empirically this smuggles in uncontrolled market beta.

Design notes (from the strategy's source file).

Aggressive-sizing variants of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.006
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	raw
cap_mode	priority
beta_overlay	0.0

Performance

metric	s7 directional aggressive	market (CRSP VW)
Annualized return	11.78%	9.94%
Annualized volatility	11.85%	19.35%
Sharpe (NAV-pct)	1.00	0.59
Sortino	1.45	
Max drawdown	-33.0%	-55.5%
Calmar	0.36	
Final NAV ($10M start)	$87,451,313	
Worst day	-7.17%	
% positive days	51.9%	
Sharpe 1996-2001	1.64	
Sharpe 2002-2007	1.65	

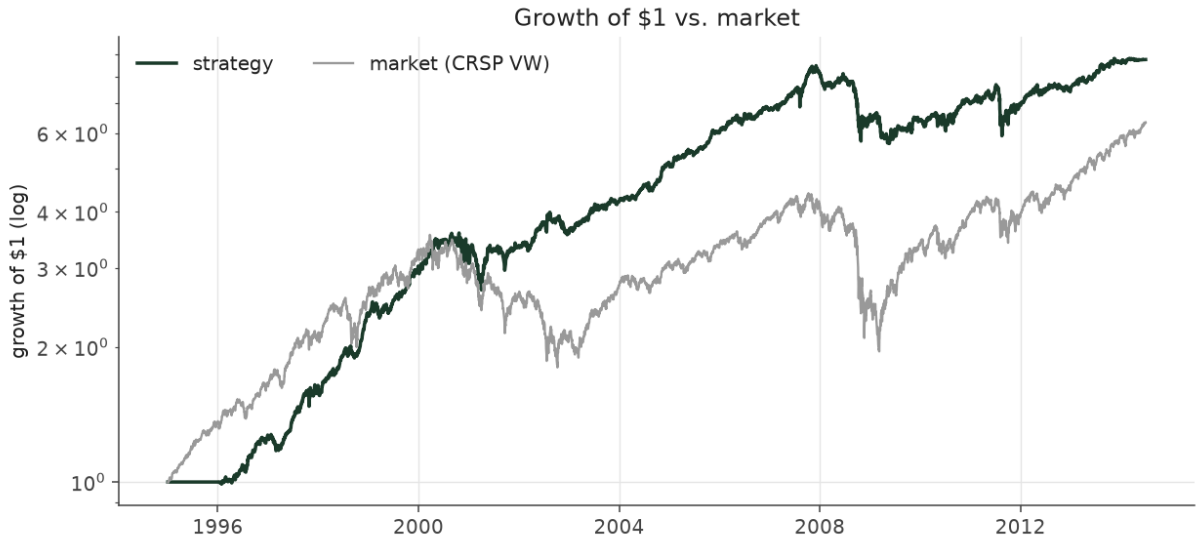
Sharpe 2008-2013	0.13	

Correlation of daily returns with the market: **0.66**.

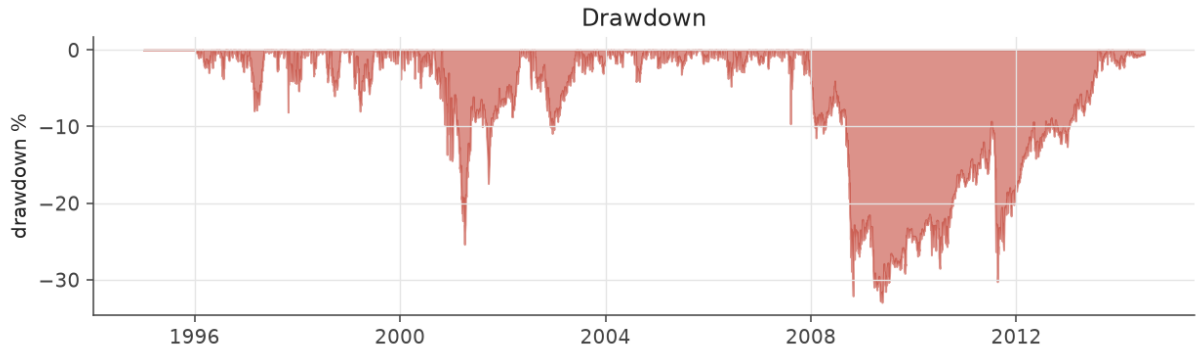
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 0.999

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.118

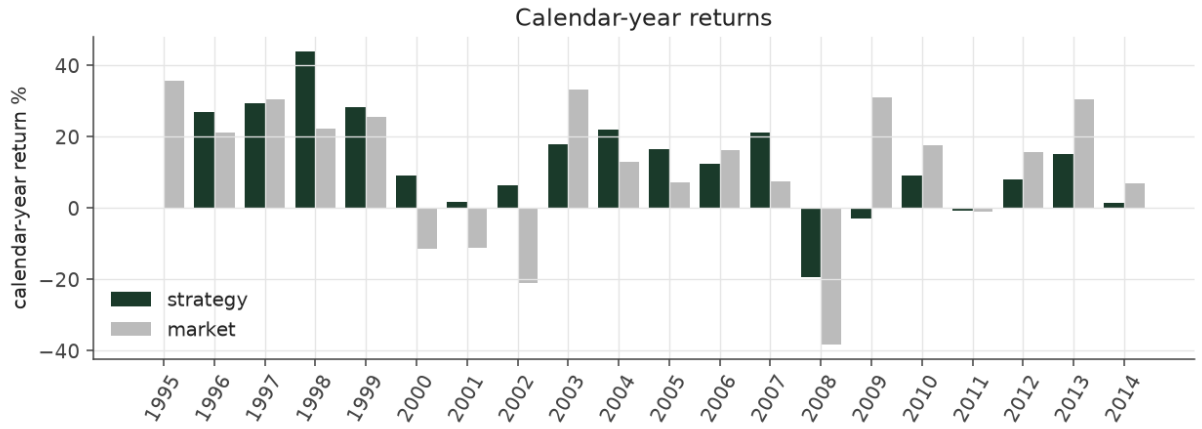
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.357



NAV (log scale)



Drawdown from running peak

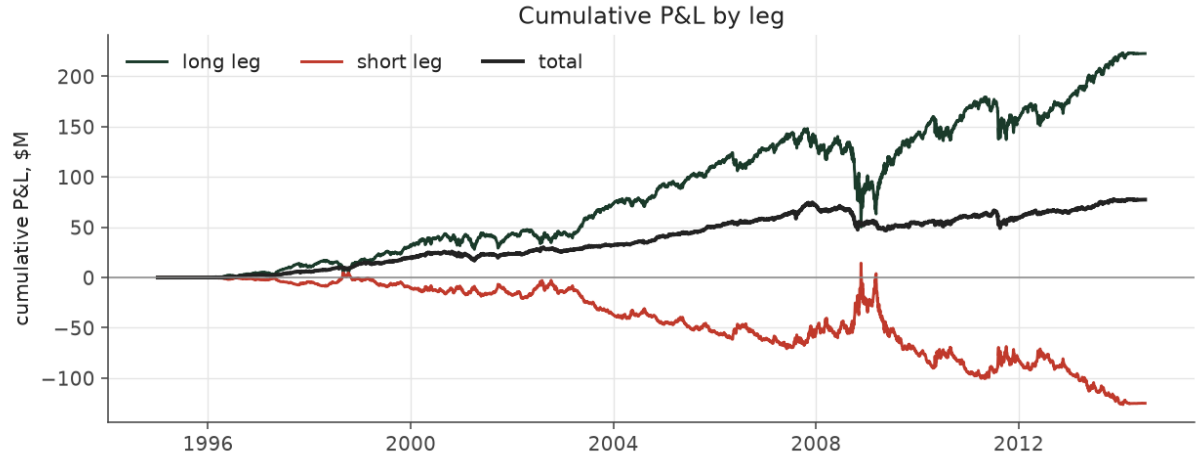


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	222.73	-124.99
P&L after costs ($M)	212.73	-135.28
of which common (average-event) component ($M)	88.48	-80.08

of which excess / PEAD component ($M)	20.72	48.19
excess minus costs ($M)	10.72	37.90
gross return per $ traded (bp)	370.11	-222.37
positions in the money (hit rate)	0.63	0.42
stand-alone Sharpe (leg only)	0.73	-0.46

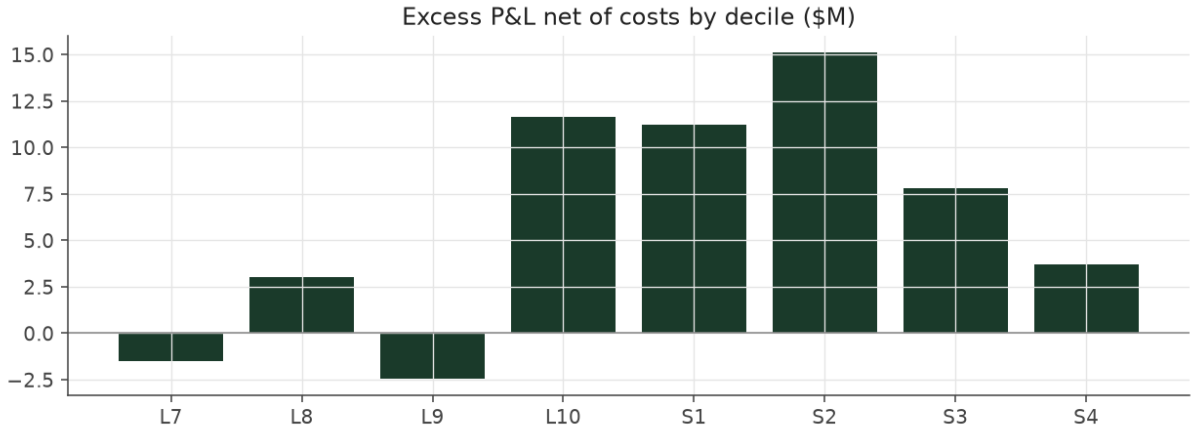
Daily P&L correlation between the two legs: **-0.97**.

• **The short leg loses money in absolute terms** (-125.0M before costs) while the long leg earns +222.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 88.5M / 80.1M. Stripping that common component, the surprise signal itself earned +20.7M on the long side and +48.2M on the short side.

• The two legs' daily P&L are -0.97 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.73 long / -0.46 short vs 1.00 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	514.45	18.09	-0.60	0.88	-1.49	-3.1%	-28.94
long	8	16,588	1,409.31	53.03	5.42	2.38	3.04	6.3%	21.59
long	9	17,747	1,906.30	56.00	0.65	3.11	-2.46	-5.1%	-12.89
long	10	17,174	2,187.84	85.60	15.25	3.63	11.62	23.9%	53.13
short	1	17,169	1,947.43	-49.31	14.87	3.63	11.24	23.1%	57.74
short	2	17,281	1,766.20	-41.46	18.29	3.18	15.11	31.1%	85.57
short	3	15,970	1,363.24	-31.22	10.33	2.50	7.83	16.1%	57.43
short	4	9,171	543.72	-13.28	4.70	0.98	3.71	7.6%	68.25



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	527.66	4.45	7.78	1.84	5.93	12.2%	112.39
2	22,885	1,725.66	9.74	17.71	4.49	13.22	27.2%	76.61

3	25,501	2,640.37	17.24	19.90	5.21	14.70	30.2%	55.66
4	27,583	3,267.00	18.85	13.42	4.94	8.48	17.4%	25.96
5	28,947	3,477.79	27.17	10.10	3.80	6.29	12.9%	18.10

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	300.19	6.42	8.06	1.50	6.56	13.5%	218.65
2	23,084	1,613.35	9.98	17.20	4.84	12.36	25.4%	76.61
3	25,851	2,851.47	15.67	24.23	5.70	18.53	38.1%	64.97
4	27,665	3,422.28	16.70	10.57	4.79	5.78	11.9%	16.90
5	28,874	3,450.23	28.63	8.78	3.45	5.33	11.0%	15.43

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	4,666.77	-12.92	16.61	7.52	9.09	18.7%	19.48
41-60d	25,652	2,632.78	22.99	11.89	4.64	7.25	14.9%	27.52
61-90d	35,324	3,344.89	17.09	18.76	6.09	12.67	26.1%	37.88
>90d	12,298	994.04	50.29	21.65	2.04	19.61	40.3%	197.29

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	1,718.35	23.31	21.02	3.00	18.02	37.1%	104.88
2002-07	39,956	4,116.20	43.81	38.63	7.05	31.59	65.0%	76.74
2008-13	45,278	5,803.94	10.33	9.25	10.24	-0.99	-2.0%	-1.70

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	1,238.86	20.07	5.99	2.33	3.67	7.5%	29.58
Chems	2,590	701.65	6.49	6.27	1.05	5.22	10.7%	74.43
Durbl	2,843	766.16	-0.97	1.37	1.16	0.21	0.4%	2.73
Enrgy	6,194	937.74	5.51	12.17	1.50	10.67	21.9%	113.80
Hlth	11,696	1,162.56	14.23	7.15	2.13	5.02	10.3%	43.16
Manuf	12,034	1,066.72	5.86	5.38	1.91	3.47	7.1%	32.52
Money	17,583	1,062.83	6.91	9.58	2.08	7.50	15.4%	70.58
NoDur	5,081	882.91	8.96	4.94	1.43	3.51	7.2%	39.72
Other	18,377	1,189.85	8.11	8.34	2.35	6.00	12.3%	50.40
Shops	11,984	1,123.54	9.69	7.24	2.00	5.24	10.8%	46.62

Telcm	3,981	796.70	-7.22	-0.46	1.21	-1.67	-3.4%	-20.93
Utils	4,672	708.98	-0.19	0.94	1.15	-0.21	-0.4%	-2.96

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	3,964.97	9.35	14.03	6.36	7.67	15.8%	19.35
low vol	40,091	3,751.90	35.21	29.62	7.09	22.52	46.3%	60.04
mid vol	37,903	3,786.09	32.34	25.14	6.57	18.57	38.2%	49.05

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	11.78%	0.00%
0.5%	5.5	11.40%	0.38%
1.0%	11.1	11.00%	0.78%
2.0%	22.2	10.11%	1.67%
5.0%	55.4	6.16%	5.62%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.